

Asymptotic-Safety Truncations as Conditional Shadows: Fixed-Point Residuals, Error Bounds, and Scheme Dependence

Peter Nero

July 2026
Version 2

Abstract

Finite functional-renormalization-group truncations can reveal fixed-point structure, but a truncation fixed point is not automatically the shadow of an exact ultraviolet completion. This paper gives a validation framework for that inference. A contraction residual bounds the distance to an exact fixed point when the exact map is already known to be contractive. A finite truncation has its own fixed point only when an independent existence condition, such as invariant contraction or a verified degree argument, is supplied. If a lifted truncation fixed point has a certified defect relative to the exact map, its distance from the exact fixed point is bounded by that defect divided by the contraction margin. An additive-error Lipschitz inequality alone proves none of these conclusions. Applied to Modal Triplet Theory, the framework identifies the missing data: a selected map from coherent variables to FRG couplings, a physical regulator and scheme, and certified truncation defects. The paper therefore supplies a rigorous truncation diagnostic, not a proof that current asymptotic-safety truncations are MTT predictions.

Version 2 Revision Note

Supersedes	Version 1.0, <i>Asymptotic Safety as a Truncation Shadow of a Coherent-Sector UV Endpoint</i> .
Reason	The previous argument inferred approximate fixed points from an additive-error contraction and treated truncation convergence as if an exact fixed point and coupling map had already been established.
Resolution	The paper now separates exact-map contraction, finite-map existence, lifted residuals, and scheme/truncation error. A counterexample records why the former inference fails.
Retained result	Under a genuine contraction and a certified lifted defect, truncation fixed points obey an explicit a posteriori error bound.
Remaining boundary	No selected MTT-to-FRG coupling chart or certified physical truncation defect is currently available. Numerical FRG fixed points remain external diagnostics at this tier.

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1 Why a separate truncation paper is needed

The technical MTT-asymptotic-safety paper owns the conditional conjugacy and quasi-compact unstable-subspace theorems. This companion paper addresses a narrower question:

When does a fixed point of a finite FRG truncation
approximate a fixed point of an exact map?

This question matters because practical asymptotic-safety calculations replace an infinite-dimensional effective action by finitely many couplings. Stability across larger truncations and regulator choices is valuable evidence [3], but the logical implication to an exact fixed functional requires error control.

2 Exact map, truncation, and lift

Let $\mathcal{X}$ be a Banach space, $\mathcal{K} \subset \mathcal{X}$ a nonempty closed set, and

$$\mathcal{T} : \mathcal{K} \rightarrow \mathcal{K}$$

the exact dimensionless RG step. For each N , let X_N be a finite-dimensional coupling space, with

$$\pi_N : \mathcal{X} \rightarrow X_N, \quad \iota_N : X_N \rightarrow \mathcal{X}.$$

The finite map

$$\mathcal{T}_N : K_N \rightarrow K_N$$

need not equal $\pi_N \mathcal{T} \iota_N$. Gauge fixing, regulator choice, projection, closure assumptions, and numerical discretization can all contribute to the defect.

Definition 2.1 (Lifted defect). For $y \in K_N$, define

$$d_N(y) = \|\mathcal{T}(\iota_N y) - \iota_N \mathcal{T}_N(y)\|_{\mathcal{X}}.$$

This is the error relevant to fixed-point transfer.

Convergence of individual couplings or visual stability of a plot is not a substitute for a bound on d_N in a declared norm.

3 Residual validation

Theorem 3.1 (A posteriori fixed-point bound). *Suppose $\mathcal{T} : \mathcal{K} \rightarrow \mathcal{K}$ is a contraction with constant $0 \leq q < 1$, and let x_* be its unique fixed point. Then every $x \in \mathcal{K}$ satisfies*

$$\|x - x_*\|_{\mathcal{X}} \leq \frac{\|\mathcal{T}x - x\|_{\mathcal{X}}}{1 - q}.$$

Proof. Using $\mathcal{T}x_* = x_*$,

$$\|x - x_*\| \leq \|x - \mathcal{T}x\| + \|\mathcal{T}x - \mathcal{T}x_*\| \leq \|\mathcal{T}x - x\| + q\|x - x_*\|.$$

Rearrange. □

The theorem is a validation result, not an existence result from a small residual. Existence and uniqueness came from the exact contraction hypothesis.

Corollary 3.2 (Certified truncation shadow). *Assume the hypotheses of [Theorem 3.1](#). If $\mathcal{T}_N y_N = y_N$, $\iota_N y_N \in \mathcal{K}$, and*

$$d_N(y_N) \leq \varepsilon_N,$$

then

$$\|\iota_N y_N - x_*\|_{\mathcal{X}} \leq \frac{\varepsilon_N}{1 - q}.$$

Consequently, $\varepsilon_N \rightarrow 0$ implies convergence of the lifted truncation fixed points to x_ .*

Proof. At a truncation fixed point,

$$\|\mathcal{T}(\iota_N y_N) - \iota_N y_N\| = \|\mathcal{T}(\iota_N y_N) - \iota_N \mathcal{T}_N y_N\| = d_N(y_N).$$

Apply [Theorem 3.1](#). □

4 The finite fixed point needs its own theorem

Proposition 4.1 (Finite-map existence by contraction). *If K_N is complete and $\mathcal{T}_N : K_N \rightarrow K_N$ is a contraction with constant $q_N < 1$, then $\mathcal{T}_N$ has a unique fixed point y_N .*

This is one available route. In finite dimension, an invariant compact convex set and continuity permit Brouwer existence, but not uniqueness or the a posteriori error bound above. Interval Newton, Krawczyk, degree, or Conley-index methods can provide stronger computer-assisted certificates.

4.1 Why additive-error contraction is insufficient

The inequality

$$\|\mathcal{T}_N x - \mathcal{T}_N y\| \leq q\|x - y\| + \varepsilon, \quad q < 1, \quad \varepsilon > 0, \tag{1}$$

does not make $\mathcal{T}_N$ a contraction.

Example 4.2 (No existence from (1)). Let $K = \{0, 1\}$ with its usual metric and let T exchange the two points. Then K is complete, $T : K \rightarrow K$ has no fixed point, and

$$|Tx - Ty| \leq 0|x - y| + 1$$

for all $x, y \in K$. Thus (1) can hold with $q = 0$ without a fixed point.

This is the precise failure in the earlier version. An additive defect can be useful after an exact or finite fixed point has been established, but it cannot replace the existence theorem.

5 Perturbing an exact contraction

Theorem 5.1 (Fixed-point perturbation). *Let $T, S : \mathcal{K} \rightarrow \mathcal{K}$ be contractions, with T having constant $q < 1$, and fixed points x_T, x_S . If*

$$\sup_{x \in \mathcal{K}} \|Sx - Tx\| \leq \varepsilon,$$

then

$$\|x_S - x_T\| \leq \frac{\varepsilon}{1 - q}.$$

Proof.

$$\|x_S - x_T\| \leq \|Sx_S - Tx_S\| + \|Tx_S - Tx_T\| \leq \varepsilon + q\|x_S - x_T\|.$$

□

The theorem requires S to possess a fixed point, here guaranteed by its own contraction property. Closeness to T alone is not enough.

6 Scheme and regulator dependence

An FRG truncation is specified by more than a list of operators. It includes:

- field parametrization and background split;
- gauge fixing and ghost sector;
- regulator shape R_k ;
- projection prescriptions for beta functions;
- truncation basis and normalization conditions;
- spacetime domain and boundary conditions.

Two calculations can be compared by treating one step map as a perturbation of the other and applying [Theorem 5.1](#) when a common invariant domain and contraction estimate exist. Without those data, “scheme stability” means empirical robustness across the tested schemes, not a theorem of universal equality.

7 Relevant directions and spectral error

Let $L = D\mathcal{T}(x_*)$ and $L_N = D(\iota_N \mathcal{T}_N \pi_N)$ at compatible points. Convergence of fixed points does not alone imply convergence of critical exponents. One needs an operator-norm or resolvent estimate and a spectral separation condition. For an isolated spectral cluster enclosed by a contour γ , the relevant comparison is between Riesz projectors

$$P = \frac{1}{2\pi i} \oint_{\gamma} (z - L)^{-1} dz, \quad P_N = \frac{1}{2\pi i} \oint_{\gamma} (z - L_N)^{-1} dz.$$

If the resolvent perturbation is small uniformly on γ , standard spectral perturbation theory controls the cluster and its total algebraic multiplicity [\[4\]](#). Counting positive critical exponents in a finite matrix without such control does not prove the exact number of relevant directions.

8 Application contract for MTT

For an MTT truncation to qualify under [Theorem 3.2](#), the following objects must be emitted from one selected branch:

1. an exact or controlled upper RG map $\mathcal{T}$;
2. a chart from coherent geometric variables to FRG actions and couplings;
3. a compatible finite basis and lift (π_N, ι_N) ;
4. an invariant domain and a contraction or other existence certificate;
5. a computable lifted defect d_N ;
6. a regulator, gauge, and normalization convention;
7. a spectral error certificate for claimed relevant directions.

The present MTT corpus does not yet provide this packet. It has exact finite internal operators, conditional four-dimensional filters, and several finite-domain constructive results. Those are ingredients, not the selected FRG coupling map or truncation certificate.

9 Status table

Statement	Status	Required input
Residual-to-distance bound	Exact	Exact contraction and a residual.
Lifted truncation convergence	Exact	Fixed points plus $d_N \rightarrow 0$.
	conditional	
Finite fixed-point existence	Separate	Contraction, degree, or validated numerical theorem.
Additive-error contraction implies a fixed point	False	Counterexample given above.
Critical-exponent convergence	Conditional	Resolvent bound and isolated spectral cluster.
Current FRG fixed points are MTT shadows	Open	Selected coupling chart and defect certificate.

10 Discussion and conclusion

The phrase “truncation shadow” is justified only when there is something precise casting the shadow: an exact map, a lift, and a controlled defect. Under those conditions, [Theorem 3.2](#) gives a clean quantitative statement. Without them, a stable truncation fixed point remains meaningful FRG evidence but not an MTT derivation.

This correction does not weaken the research program. It turns a broad correspondence into a reproducible target. The next useful computation is not another unconstrained search for fixed-point coordinates. It is a same-source construction of the MTT-to-FRG chart and a certified residual for one explicitly declared truncation.

Computational Evidence and Reproducibility

The truncation-residual and scheme-dependence results in this paper are analytic statements under displayed hypotheses. The mapped strict-upgrade ledger does not prove an exact ultraviolet completion; it records a separate open source-selection obligation.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).

Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] C. Wetterich, “Exact evolution equation for the effective potential,” *Physics Letters B* **301** (1993), 90–94.
- [2] M. Reuter, “Nonperturbative evolution equation for quantum gravity,” *Physical Review D* **57** (1998), 971–985.
- [3] M. Reuter and F. Saueressig, *Quantum Gravity and the Functional Renormalization Group*, Cambridge University Press, 2019.
- [4] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1995.
- [5] P. Nero, *Modal Triplet Theory and Asymptotic Safety: A Conditional FRG Conjugacy and Unstable-Subspace Theorem*, current revised MTT paper corpus, 2026.